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# Cooperative Pretraining for Robust Emergent Communication Under Adversarial Pressure

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## Abstract

Emergent communication in multi-agent reinforcement learning (MARL) degrades in mixed cooperative-competitive environments with public communication due to adversarial pressure [15]. This reduces the efficacy of cooperative behavior in these environments, making MARL approaches less effective. We investigate whether utilizing a cooperative pretraining stage during initial training increases the robustness of cooperative communication in adversarial environments.

To study this, we consider a partially observable multi-agent setting in which two cooperative teams compete, and all inter-agent messages are publicly observable. Using modified Multi-Agent Particle Environments, we compare agents trained directly in the adversarial environment against those that first develop communication protocols in a protected cooperative phase. We evaluate both task performance and the properties of the learned communication protocols, including the causal influence of communication on agent behavior.

## 1 Introduction

Multi-agent reinforcement learning (MARL) has shown strong potential for solving complex tasks involving multiple agents acting in a shared environment, including those with the need for coordination among agents [4]. In particular, cooperative MARL has been successful in settings where agents must learn to communicate and align their behavior to achieve common objectives. However, prior work has shown that this cooperative behavior can degrade when agents are placed under adversarial pressure in mixed cooperative-competitive environments [15]. In such settings, the presence of competing agents can disrupt communication and reduce the overall robustness of learned cooperative strategies.

This paper investigates whether a cooperative pretraining stage can improve the resilience of learned cooperation in multi-agent adversarial environments. More specifically, we examine whether first training agents in a purely cooperative setting allows them to develop communication and coordination strategies that remain more effective when later exposed to adversarial conditions.

This problem is relevant because improving robustness in adversarial MARL settings could expand the applicability of reinforcement learning systems in domains where communication security is critical. By exploring ways in which MARL environments could be made more robust to adversarial pressure, it unlocks further applications of reinforcement learning (and MARL in particular) in domains where adversarial agents or attacks are critical to circumvent.

## 2 Related Work

Lowe et al. [10] provide foundational work for modern multi-agent reinforcement learning in mixed cooperative-competitive settings, utilizing a modified actor-critic approach to MARL that formed an

important basis for later work in coordination and cooperation. Subsequent work has shown that these learned communication strategies can become fragile under adversarial pressure. Vanneste et al. [15] found that in a predator-prey setting, predator agents with public communication performed worse than agents whose communication was private, with public communication yielding performance roughly comparable to settings in which the predators could not communicate at all. Similarly, Blumenkamp et al. [1] show that when a self-interested agent is introduced into an environment with shared communication channels, the performance of the cooperative agents decreased significantly. This broader vulnerability of inter-agent communication is also supported by Tu et al. [14], who show that adversarial messages can severely degrade performance in cooperative multi-agent systems. Likewise, Xue et al. [7], demonstrate that cooperative MARL policies can be effectively disrupted by adversarial policies.

Our approach differs from prior related work by focusing not only on how cooperation deteriorates under adversarial pressure, but whether deterioration can be mitigated through training design. While Vanneste et al., Blumenkamp et al., Tu et al., and Xue et al. primarily show that communication and coordination become vulnerable when adversarial agents can observe or exploit them, we will investigate whether a cooperative pretraining stage can improve robustness before agents enter adversarial settings. We therefore build on these findings by asking not just whether adversarial pressure harms cooperation, but whether that harm can be reduced through prior cooperative learning.

### 3 Problem Statement

Let  $\mathcal{A} = \mathcal{C} \cup \mathcal{E}$  denote the full set of agents, where  $\mathcal{C} = \{s, l\}$  is a cooperative team consisting of a speaker  $s$  and a listener  $l$ , and  $\mathcal{E} = \{e\}$  is an adversarial eavesdropper. At each timestep  $t$ , the speaker observes a target state  $x \in \mathcal{X}$  and emits a discrete message  $m \in \mathcal{M}$ , where  $\mathcal{M} = \Sigma^L$  is the space of messages of length  $L$  over a vocabulary  $\Sigma$  with  $|\Sigma| = V$ . The message is transmitted over a *public* channel observable by all agents in  $\mathcal{A}$ .

The listener receives  $m$  and must identify the target  $x$  from a distractor set  $\mathcal{D} = \{x, d_1, \dots, d_k\}$ , producing a prediction  $\hat{x} = \pi_l(m)$ . The adversary simultaneously observes  $m$  and attempts to reconstruct  $x$ , producing  $\tilde{x} = \pi_e(m)$ .

The cooperative team’s objective is to maximize listener accuracy while minimizing information leakage to the adversary:

$$\max_{\pi_s, \pi_l} \mathbb{E}[\mathbf{1}[\hat{x} = x]] - \lambda \cdot \mathbb{E}[\mathbf{1}[\tilde{x} = x]] \quad (1)$$

where  $\lambda \geq 0$  is a penalty coefficient balancing task performance against information leakage. The adversary’s objective is the converse:

$$\max_{\pi_e} \mathbb{E}[\mathbf{1}[\tilde{x} = x]] \quad (2)$$

This constitutes a zero-sum tension embedded within a cooperative communication task: the cooperative team must develop a signaling protocol that is simultaneously *informative* to the listener and *opaque* to the eavesdropper, despite sharing the same public channel.

The central question of this work is whether the trajectory through policy space taken during training affects the team’s ability to reach such an equilibrium. Formally, let  $\alpha \in [0, T]$  denote the number of timesteps devoted to cooperative pretraining (phase 1) before adversarial agents are introduced (phase 2), with  $\alpha = 0$  corresponding to the simultaneous baseline. We ask: for what values of  $\alpha > 0$  do the resulting policies  $(\pi_s, \pi_l)$  achieve strictly higher cooperative task performance and lower information leakage than the  $\alpha = 0$  baseline?

### 4 Method

Emergent communication in MARL degrades under adversarial pressure when messages are transmitted over public channels, as predicted by game-theoretic analysis [9, 6]. Simultaneously trained agents must navigate the tension between the incentive to inform the partner and the cost of leaking information to the adversary from the outset. Our key insight is that this failure is likely a learning dynamics problem, not an equilibrium instability, because classical cheap-talk framework fails to generalize to environments where one audience has an asymmetric shared prior with the speaker,

allowing for asymmetric posterior belief extraction which cannot be represented in the theoretical framework posed by Farrell and Gibbons, 1989 [6].

Abadi and Andersen (2016) [8] demonstrate that this kind of emergent coding arises naturally in neural network systems without a prescribed cryptographic algorithm. Yu, et al., (2022) [3] extend this to emergent communication specifically, showing that speaker-listener coalitions can develop covert signaling protocols in adversarial reference games, but only when granted asymmetric advantages over the adversary.

To this end, our agent architecture uses actor-critic networks trained with MADDPG [10], which provides centralized critics during training while permitting decentralized execution. The communication module uses Gumbel-Softmax relaxation [2] to enable gradient-based optimization of discrete messages during training, with argmax discretization at evaluation.

The core method is a two-phase training pipeline inspired by curriculum learning principles [16]. During phase one the cooperative team trains in isolation on the referential communication task. No adversarial agents are present and the messages are private, so the training objective is simply to maximize the listener’s accuracy. In phase 2, the adversarial eavesdropper is introduced, the communication channel becomes public, and all agents train simultaneously. The cooperative team’s policies, including the communication module, continue to update. This phase tests both whether the established conventions robustly resist the inhibition effect [6] and whether the shared context from pretraining enables the cooperative team to sustain a coded separating equilibrium.

The transition between phases is, for now, instantaneous, with each stage running for  $\alpha$  and  $T - \alpha$  respectively, where  $\alpha = 0$  is the baseline simultaneous approach. The instantaneous transition is a simplification, and future work will extend the framework to characterize the behavior of gradually transitioned pipelines. To supplement this for now, we will log performance and checkpoints at fine-grained intervals around the transition point for coarse analysis.

## 5 Experimental Design

### Hypotheses

1. **Task Performance:** Cooperatively pretrained agents will achieve higher task performance than simultaneously trained agents when deployed in an adversarial environment with public communication
2. **Communication Robustness:** Communication protocols established during cooperative pretraining will exhibit graceful degradation under adversarial pressure, consistent with Crawford and Sobel’s partition equilibrium predictions, rather than the protocol collapse observed in simultaneously trained agents.
3. **Information Leakage:** Pretrained cooperative agents will leak less actionable information to adversarial eavesdroppers than simultaneously trained agents, as measured by adversary task accuracy.

### Domain

We conduct all experiments in modified Multi-Agent Particle Environments (MPE) implemented via PettingZoo [13]. MPE is a standard 2D multi-agent benchmark used extensively in emergent communication and mixed-motive MARL research [10, 5]. We modify the `simple_crypto` environment to support a team-based public-eavesdropping setting mimicking the cheap-talk setting analyzed in previous work [6]:

- **Cooperative Team:** A speaker agent observes a target state composed of structured attributes and sends a discrete message to a listener agent, who must identify the correct target from a set of distractors. The team is rewarded for listener accuracy and penalized for information leaked to the adversary.
- **Adversary:** An eavesdropper agent observes the cooperative team’s messages on a public channel and attempts to reconstruct the target state. The adversary is rewarded for reconstruction accuracy.

- **Communication Channel:** Messages are discrete sequences of 3 symbols drawn from a vocabulary of size 10. The vocabulary and message length are deliberately constrained to create a communication bottleneck that encourages compositional protocol structure [12]. All messages are publicly observable

## Experiments

We use a two-stage screen-and-confirm procedure. In the screening stage, we train 5 seeds across six pretraining phase sizes  $\in \{1000, 5000, 10000, 15000, 20000, 30000\}$  followed by 100,000 public steps to characterize how pretraining duration affects outcomes and to select the best-performing configuration. We skip very high ratios as these provide insufficient adversarial time for analysis. In the confirmation stage, we train 12 fresh agents under the selected configuration and compare against the baseline. The specific number of seeds used is due to time constraints, and future work will extend the analysis with a higher sample size.

## Evaluation Criteria

1. **H1 - Task Performance:** Compare the proportion of evaluation episodes in which the listener correctly identifies the target, used to calculate an effect size compared to the baseline, as well as confidence intervals.
2. **H2 - Communication Robustness:** Causal Influence of Communication (CIC) measures whether messages causally influence the listener’s action distribution using counterfactual message substitution [11].  $CIC > 0$  confirms genuine communication rather than epiphenomenal correlation.
3. **H3 - Information Leakage:** Measure eavesdropper accuracy and the ratio of eavesdropper accuracy divided by listener accuracy.

## 6 Results and Discussion

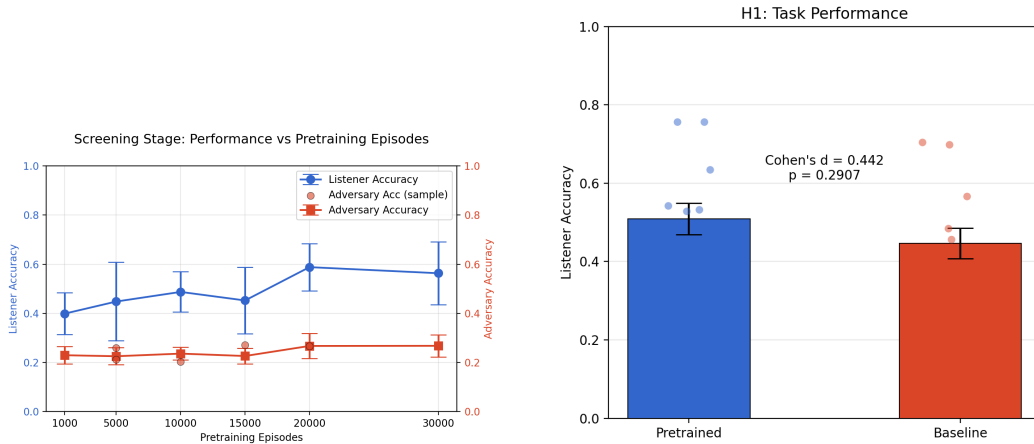


Figure 1: Overall comparison of listener accuracies and pretraining listener vs. adversary accuracy.

From the experiments described above, we find that cooperative pretraining has a modest positive effect in mixed cooperative-competitive MARL environments. As shown in Figure 1, our method achieved a 7% improvement in listener accuracy during the confirmation stage. Notably, we find a correlation between listener accuracy and adversary accuracy within the baseline that is not present in the intervention, with the intervention adversary accuracy staying indistinguishable from random noise.

We additionally find that the performance of the agents appears to split into two different regimes: low performance noisy communication with low CIC, and high-listener-performance. This phenomenon appears under both training methods, but cooperative pretraining increases the proportion of seeds

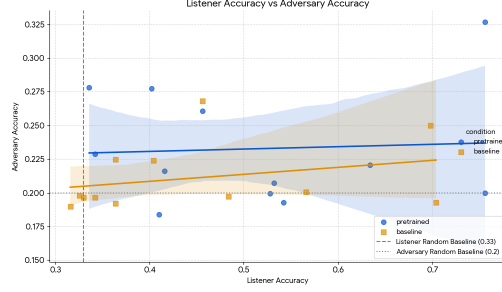


Figure 2: Correlation Plot for Listener Accuracy and Adversary Accuracy for Pretrained and Baseline conditions. Pretrained = 0.058, Baseline = 0.286. Note that more runs would be valuable for better confidence here.

that converge to the high-performance regime. Detailed analysis of this phenomenon is left for future work.

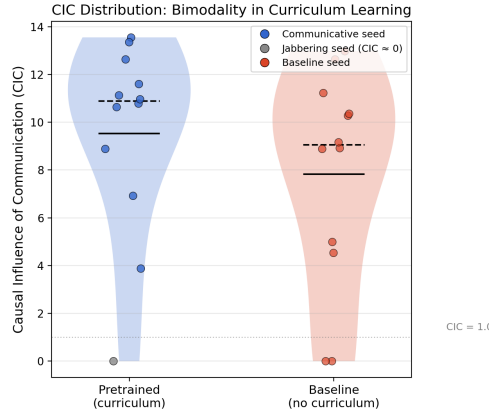


Figure 3: Violin Plot of CIC for both pretrained and baseline, showing the increased density of high-performance regime seeds in the pretrained setting.

With the important caveat that additional training runs are needed for statistical confidence, our results provide preliminary evidence consistent with hypotheses 1, 2, and 3, as listener accuracy was improved by our intervention, the adversary extracted less actionable information than in the baseline, and the density of high CIC seeds was higher within the intervention group. Future work should extend these results with more training runs to allow for more detailed and statistically significant analysis.

We also found that our results were strongly influenced by the reward structure of the environment. Specifically, the penalty coefficient controlling how much the adversary’s reward is subtracted from the cooperative reward played a major role in determining whether pretraining was useful. When this coefficient was less than 1, agents tended to converge to policies that maximized listener accuracy while largely ignoring information leakage. When it was too high, agents were penalized so heavily for informative communication that they struggled to learn a useful protocol at all. This suggests that cooperative pretraining is most effective only when the environment provides a balanced incentive between task performance and adversarial resistance.

Overall, these results indicate that cooperative pretraining can improve cooperative task performance under adversarial pressure, but it does not by itself solve the problem of public-channel information leakage. The main effect appears to be that pretrained agents enter the adversarial phase with an already established communication convention, which helps preserve listener performance. However, because the communication channel remains public, the same structure that helps the listener may also remain available to the adversary.

## 7 Conclusion and Future Work

Emergent communication in MARL environments has been shown to degrade in mixed cooperative-competitive settings, especially when communication occurs over public channels. In this project, we investigated whether a cooperative pretraining phase could improve the robustness of learned communication before agents are exposed to adversarial pressure. Our results show that cooperative agents trained with a pretraining phase perform better than agents trained directly in the adversarial environment. Since this method is a simple change to the training procedure rather than a modification to the underlying architecture, it may be useful as a general strategy for improving communication-based MARL systems.

The main discovery from this project is that cooperative pretraining appears to make emergent communication protocols more robust to adversarial eavesdropping, and appears to increase the likelihood of highly-causal communication. It also does slightly increase the listener’s accuracy, though its most significant result is communication opacity to the adversary.

Future work should explore the effects of cooperative pretraining in richer multi-step communication environments, rather than only in the one-step game used in this project. Multi-step settings may allow agents to develop more complex protocols that depend on temporal context, shared history, or sequential coordination. We also plan to examine these findings more rigorously through a larger hyperparameter sweep and additional confirmation runs with more random seeds to allow for statistical reliability. In particular, future experiments should study how pretraining duration, message length, vocabulary size, and adversary penalty coefficient affect the tradeoff between listener accuracy and adversarial leakage. Future work should also investigate the apparent multi-attractor behavior shown in Figure 3, and analyze the effects of a non-instantaneous introduction of the adversary.

## 8 Code

All code for this project is available at <https://github.com/ChristianRogers1/friends-first-marl>.

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